

DHCP, watched from both ends

Two VMs on a private LAN: one hands out addresses, the other asks for one. Every block below was captured while the lab was running — including a real DORA exchange, recorded on the server and on the client at the same instant.

Command `qlab run dhcp-lab`

VMs `dhcp-lab-server` 192.168.100.1 · `dhcp-lab-client` by DHCP

Credentials `labuser` / `labpass`

Outcome `qlab test dhcp-lab` → all exercises passed

1. What is in the lab

VM	ROLE
dhcp-lab-server 192.168.100.1	Ubuntu 22.04 running isc-dhcp-server , bound to the lab LAN only.
dhcp-lab-client address from the pool	The same image with no server, asking for an address the ordinary way.

Each VM has two network cards: the lab LAN, and the SLIRP one QLab attaches to every VM so `qlab shell` works. That second card matters more than it looks — it is why the server must be told *which* interface to serve, and it is the first thing to check when a DHCP server appears to do nothing.

```
lo                UNKNOWN    127.0.0.1/8 ::1/128
ens3              UP          10.0.2.15/24 metric 100 fec0::5054:ff:fe12:3456/64
fe80::5054:ff:fe12:3456/64
ens4              UP          192.168.100.1/24 fe80::5054:ff:fe00:301/64

/etc/default/isc-dhcp-server:INTERFACESv4="ens4"
```

`INTERFACESv4="ens4"` is not decoration. Without it `isc-dhcp-server` either refuses to start or, worse, starts serving the wrong network.

2. The server is up

```
active
 1635      02:00 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcp/
dhcpd.conf ens4
```

Its configuration is short enough to read in full:

```
option domain-name "dhcp-lab.local";
option domain-name-servers 8.8.8.8, 8.8.4.4;
default-lease-time 600;
max-lease-time 7200;
authoritative;
subnet 192.168.100.0 netmask 255.255.255.0 {
    range 192.168.100.100 192.168.100.200;
    option routers 192.168.100.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 192.168.100.255;
}
```

Four things are being declared here: the **pool** (`range`), how long an address is lent for (`default-lease-time` / `max-lease-time`), what else to tell the client besides its address (`option routers` , `option domain-name-servers`), and `authoritative` — which says "on this network, I am the DHCP server", so the server answers a client asking for a wrong address with a DHCPNAK instead of staying silent.

3. DORA, recorded from both sides

The four packets everyone learns about, made to happen on purpose: `tcpdump` runs on the server while the client is told to drop its lease and ask again.

```
--- on the server: tcpdump -i any -n port 67 or port 68 ---
06:27:27.076328 ens4 In IP 192.168.100.100.68 > 192.168.100.1.67: BOOTP/DHCP, Request from
52:54:00:00:03:02, length 300
06:27:28.123100 ens4 B IP 0.0.0.0.68 > 255.255.255.255.67: BOOTP/DHCP, Request from 52:54:00:00:03:02,
length 300
06:27:29.123479 ens4 Out IP 192.168.100.1.67 > 192.168.100.101.68: BOOTP/DHCP, Reply, length 300
06:27:29.124014 ens4 B IP 0.0.0.0.68 > 255.255.255.255.67: BOOTP/DHCP, Request from 52:54:00:00:03:02,
length 300
06:27:29.129241 ens4 Out IP 192.168.100.1.67 > 192.168.100.101.68: BOOTP/DHCP, Reply, length 300

--- on the client: dhclient -v ---
DHCPDISCOVER on ens4 to 255.255.255.255 port 67 interval 3 (xid=0x11d0d637)
DHCPOFFER of 192.168.100.101 from 192.168.100.1
DHCPREQUEST for 192.168.100.101 on ens4 to 255.255.255.255 port 67 (xid=0x37d6d011)
DHCPACK of 192.168.100.101 from 192.168.100.1 (xid=0x11d0d637)
bound to 192.168.100.101 -- renewal in 241 seconds.
```

Read the two halves together, because they disagree in an instructive way.

The client names each step: **DISCOVER**, **OFFER**, **REQUEST**, **ACK**. The capture on the server shows only `BOOTP/DHCP, Request` and `BOOTP/DHCP, Reply` — twice each. That is not `tcpdump` being unhelpful: DHCP is an extension of BOOTP, and at that level there are only requests and replies. Which of the four it is lives in DHCP **option 53**, inside the payload. The `-v` flag on `tcpdump` would print it.

Two more details worth noticing:

- The first two packets go to `255.255.255.255` from `0.0.0.0`. The client has no address yet, so it cannot send a normal unicast and cannot be answered by one. Everything before the ACK is a broadcast conversation.
- The `xid` ties the four packets into one transaction, which is how a client tells its own exchange apart from every other machine shouting on the same broadcast domain. Look closely and the REQUEST appears to carry a different one: it is the same number with its bytes reversed. That is a display quirk of `dhclient`, not a second transaction — compare the DISCOVER and the ACK, which agree.

4. The lease exists on both machines

The client now holds what it was given:

```
lo UNKNOWN 127.0.0.1/8 ::1/128
ens3 UP 10.0.2.15/24 metric 100 fec0::5054:ff:fe12:3456/64
fe80::5054:ff:fe12:3456/64
ens4 UP 192.168.100.101/24 fe80::5054:ff:fe00:302/64
```

And the server wrote it down. A lease is a record with an expiry, not an assignment:

```

    client-hostname "dhcp-lab-client";
}
lease 192.168.100.101 {
    starts 1 2026/09/14 06:26:12;
    ends 1 2026/09/14 06:27:27;
    tstp 1 2026/09/14 06:27:27;
    cltt 1 2026/09/14 06:26:12;
    binding state free;
    hardware ethernet 52:54:00:00:03:02;
}
lease 192.168.100.101 {
    starts 1 2026/09/14 06:27:29;
    ends 1 2026/09/14 06:37:29;
    cltt 1 2026/09/14 06:27:29;
    binding state active;
    next binding state free;
    rewind binding state free;
    hardware ethernet 52:54:00:00:03:02;
    client-hostname "dhcp-lab-client";
}

```

Two leases, one client

The lease file holds more than one entry for this MAC, and only the last is `binding state active`. Every `dhclient -r` hands an address back and the next DISCOVER may be answered with the same one or with the next free one — the server decides, and both are correct. An address is a loan with a deadline, not property, and the expired entries stay in the file as history.

Note `client-hostname "dhcp-lab-client"` in the record: the client sent its name in **option 12**, which is how a DHCP server can register names in DNS without anyone configuring anything by hand.

The client keeps its own copy, with the options it was handed:

```

}
lease {
    interface "ens4";
    fixed-address 192.168.100.101;
    option subnet-mask 255.255.255.0;
    option routers 192.168.100.1;
    option dhcp-lease-time 600;
    option dhcp-message-type 5;
    option domain-name-servers 8.8.8.8,8.8.4.4;
    option dhcp-server-identifier 192.168.100.1;
    option broadcast-address 192.168.100.255;
    option domain-name "dhcp-lab.local";
    renew 1 2026/09/14 06:31:30;
    rebind 1 2026/09/14 06:36:14;
    expire 1 2026/09/14 06:37:29;
}

```

5. Verification

Exercise 1 – DHCP Anatomy
 Exercise 1: All 6 checks passed
 Exercise 2 – Client Lease
 Exercise 2: All 3 checks passed

```
Exercise 3 – Server Leases
  Exercise 3: All 3 checks passed
Exercise 4 – DHCP Configuration
  Exercise 4: All 7 checks passed
Exercise 5 – Troubleshooting
  Exercise 5: All 4 checks passed
Exercises run:      5
Exercises passed:   5
Exercises failed:   0
Exercises skipped:  0
All exercises passed!
```

6. What to take away

- A DHCP server serves an **interface**, not a machine. On a host with two cards the single most common failure is serving the wrong one — or being refused because you did not say which.
- `tcpdump` shows BOOTP requests and replies; the four DORA names live in option 53 inside the packet. The client's own log is the easier place to read them.
- The whole exchange before the ACK is broadcast traffic, because the client has no address to be reached at.
- A lease is a loan with an expiry. Releasing one does not get it back, and the server is free to hand out a different address next time.
- `authoritative` changes what happens when a client asks for something wrong: a clear refusal instead of silence.

`guide.md` in the plugin takes this further — lease times, custom options, static reservations by MAC, and two pools on one subnet.